Exponential and Logarithmic Equations

ANSWER KEY

Exponential equations

1 Solve $3^{x+1} = 81$ exactly.
$3^{x+1} = 3^4$, so $x + 1 = 4$ and $x = 3$.

2 Solve $5^{2x} = 17$. Give an exact answer and a decimal to three places.
$2x = \log_5 17 = \frac{\ln 17}{\ln 5}$, so $x = \frac{\ln 17}{2\ln 5} \approx 0.880$.

3 Solve $e^{2x} - 5e^x + 6 = 0$ exactly.
Let $u = e^x$: $u^2 - 5u + 6 = (u-2)(u-3) = 0$, so $e^x = 2$ or $e^x = 3$: $x = \ln 2$ or $x = \ln 3$.

Logarithmic equations

4 Solve $\ln(2x-3) = 1$. Give an exact answer and a decimal to three places.
$2x - 3 = e$, so $x = \frac{e+3}{2} \approx 2.859$.

5 Solve $\log_2 x + \log_2 (x-6) = 4$, checking for extraneous solutions.
$\log_2 (x(x-6)) = 4$, so $x^2 - 6x - 16 = 0$, giving $(x-8)(x+2) = 0$. Reject $x = -2$ (logarithm of a negative number): $x = 8$.

6 Solve $\log(x+3) - \log(x-1) = 1$, checking for extraneous solutions.
$\frac{x+3}{x-1} = 10$, so $x + 3 = 10x - 10$ and $x = \frac{13}{9}$. Check: $x > 1$, valid.

7 The pH of a solution is $\text{pH} = -\log[\text{H}^+]$. Find the hydrogen-ion concentration $[\text{H}^+]$ of a solution with pH 4.6, in scientific notation to two significant figures.
$[\text{H}^+] = 10^{-4.6} \approx 2.5 \times 10^{-5}$ mol/L.

More exponential equations

8 Solve $2^x = 3^{x-1}$, giving an exact expression and a decimal to three places.
Take logs: $x\ln 2 = (x-1)\ln 3$, so $x(\ln 2 - \ln 3) = -\ln 3$, giving $x = \frac{\ln 3}{\ln 3 - \ln 2} = \frac{\ln 3}{\ln(3/2)} \approx \frac{1.0986}{0.4055} \approx 2.710$.

9 Solve $4^x - 3 \cdot 2^x - 4 = 0$ exactly, using the substitution $u = 2^x$.
$4^x = (2^x)^2 = u^2$: $u^2 - 3u - 4 = 0$, so $(u-4)(u+1) = 0$. $u = 4$ (rejecting $u = -1$, since $2^x > 0$): $2^x = 4$, so $x = 2$.

Systems and applications

10 A bacteria culture grows according to $N(t) = 500e^{kt}$. If the population reaches 2000 after 6 hours, find k (4 d.p.) and hence the time for the population to reach 10,000, to one decimal place.

$2000 = 500e^{6k}$: $e^{6k} = 4$, so $k = \frac{\ln 4}{6} \approx 0.2310$. For $N = 10000$: $10000 = 500e^{0.2310t}$, so $e^{0.2310t} = 20$, $t = \frac{\ln 20}{0.2310} \approx 12.97 \approx 13.0$ hours.

11 Solve the equation $e^x + e^{-x} = 3$ exactly, using the substitution $u = e^x$ (this expression is related to $2\cosh x$).

Multiply by $u = e^x$: $u^2 - 3u + 1 = 0$, so $u = \frac{3 \pm \sqrt{5}}{2}$. Both roots are positive, giving $x = \ln\left(\frac{3+\sqrt{5}}{2}\right)$ or $x = \ln\left(\frac{3-\sqrt{5}}{2}\right)$.

Combined logarithmic and exponential practice

12 Solve $\log_3(x+2) + \log_3(x-2) = 2$, checking for extraneous solutions.

$\log_3((x+2)(x-2)) = 2$: $x^2 - 4 = 9$, so $x^2 = 13$, $x = \pm\sqrt{13}$. Reject $x = -\sqrt{13}$ (makes $x - 2 < 0$): $x = \sqrt{13}$.

13 Solve $9^x = 27^{x-1}$ exactly, writing both sides with base 3.

$3^{2x} = 3^{3(x-1)}$: $2x = 3x - 3$, so $x = 3$.